



Lima Megacity's Influence on Aquatic Microbial Communities in the Rímac River: Dominance Over Spatial and Seasonal Variations

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Abstract

The Rímac River, a vital watershed on the Peruvian coast, is confronted with substantial environmental challenges stemming from intensive exploitation and widespread contamination. As the primary source of water for Lima, supplying approximately 80% of the city's needs, the river is heavily impacted by pollutants from domestic, hospital, industrial, and mining effluents. These contaminants introduce microbiota that pose significant public health risks. This study utilizes 16S rRNA gene metabarcoding to characterize the bacterial communities along the Rímac River, examining both spatial (upper, middle, and lower basins) and temporal (dry and rainy seasons) variations. Over a year-long sampling period, DNA sequencing revealed pronounced microbiological differences between the Metropolitan and Regional zones, primarily driven by anthropogenic activities. Key findings include a significant reduction in microbial diversity and an increase in pathogenic bacteria within the Metropolitan zone, while the influence of seasonal variations and altitudinal gradients was comparatively minor. Betaproteobacteria emerged as the most abundant class across most samples. Notably, *Aliarcobacter cryaerophilus*, an indicator of fecal contamination and a potential public health hazard, was predominantly detected in the Metropolitan zone. These results underscore the necessity for comprehensive monitoring of the Rímac River's microbiota, incorporating advanced molecular techniques to effectively track and mitigate pollution. The study emphasizes the urgent need for robust water quality management strategies to protect this critical resource, ensuring the health and sustainability of Lima and its surrounding regions.

Keywords Metabarcoding · 16S · Nanopore · Contamination · Anthropic impact

Introduction

The Rímac River represents one of the most critical watersheds along the Peruvian coast, serving as a primary source of water for the populations of Lima and Callao [1]. However, escalating anthropogenic activities have led to intensified exploitation and contamination, resulting in severe environmental degradation of the river's aquatic ecosystem. The Rímac River supplies approximately 80% of the water consumed by the Metropolitan area of Lima, yet it concurrently receives discharges of domestic and hospital wastewater, industrial effluents, and mining tailings. These pollutants introduce a distinct microbiota, posing potential public health risks to the urban population. Investigations

conducted by the National Water Authority [1] identified 1185 contamination points across the river basin, with 260 (22%) located in the upper basin, 336 (28%) in the middle basin, and 589 (50%) in the lower basin, where Lima is situated. This extensive pollution has precipitated detrimental ecological imbalances and a decline in water flow, exacerbating the vulnerability of the river system [2].

Microbiological contamination in the Rímac River has been the focus of extensive monitoring over the years, with contributions from various institutions, including ANA, SEDAPAL, and OEFA [1]. These entities are committed to preserving this critical water resource by investigating environmental stressors across the entire basin [1] and across different seasonal conditions. Historically, many of these studies have relied on conventional culture-based microbiological methods, which are limited by their slow processing times and inability to assess unculturable microbes, thus failing to comprehensively address the river's complex contamination issues [3]. In recent years, however, the advent

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of advanced, rapid, and less labor-intensive molecular techniques has transformed the field of aquatic microbiology, enabling more reliable and versatile analyses through portable and high-throughput technologies [3].

High throughput sequencing has proven to be an invaluable tool for understanding the effects of urban pollution on aquatic environments [4–7]. Numerous studies have shown that rivers are contaminated by various sources, including agricultural runoff [8, 9], direct discharges from industrial waters [10, 11], and domestic wastewater [5, 7, 12]. Domestic wastewater, in particular, has been identified as a significant source of pathogens, which can alter the composition of bacterial communities and degrade water quality downstream. In Peru, the first comprehensive effort to characterize the microbiota of the Rímac River was conducted by Romero et al. [5], who utilized 16S rRNA gene sequencing and Illumina technology during the summer of the year 2020. Their findings revealed reduced microbial diversity in the Metropolitan zone (Lower Rímac), dominated by *Arcobacter cryaerophilus*, an emerging pathogen of public health concern.

In this study, we employ 16S rRNA gene metabarcoding to characterize the microbial communities along the Rímac River, incorporating both spatial and temporal dimensions by comparing dry and rainy seasons. We hypothesize that the microbial disparities observed by Romero et al. [5] in the Metropolitan zone will persist over time due to sustained anthropogenic pressures, while seasonal variations will naturally influence the river's microbiota. This research highlights the critical need for advanced molecular techniques in monitoring pollution in the Rímac River, offering essential data to inform strategies for the protection and sustainable management of this vital natural resource for Peru.

Material and Methods

Sampling Methodology [13, 14]

The Rímac River exhibits a bimodal seasonal pattern driven by precipitation. The rainy season extends from November to April, with peak precipitation occurring between December and March [13]. Precipitation during this season is spatially variable: the upper Rímac experiences consistent rainfall, the middle Rímac receives irregular precipitation, and the lower Rímac is characterized by low rainfall throughout the year [14]. The Department of Lima is administratively divided into provinces, with Metropolitan Lima being the most densely populated area within the Province of Lima. The Rímac River flows from east to west, traversing two provinces: Huarochirí, which is predominantly rural, and Lima, which transitions from rural landscapes in the east to the highly urbanized Metropolitan area in the west. This

work uses the terms Regional and Metropolitan to refer to these areas, respectively. Sampling was conducted across four campaigns to capture seasonal variations: August 2020 (dry season), November 2020 (rainy season), April 2021 (rainy season), and June 2021 (dry season). These campaigns spanned a full year and encompassed the entire altitudinal gradient of the Rímac River, from 4500 m above sea level (msl) at the east, to its lower reaches. A total of 15 georeferenced sampling points were selected along the river, including Chicla, San Mateo, Tamboraque, Hidroeléctrica, Chacahuaro, Matucana, Surco, Santa Eulalia, Huachipa, Chaclacayo, Libertadores, Puente Nuevo, Universitaria, Faucett, and Gambetta, with the latter five located in the Metropolitan zone (Fig. 1). Surface water samples (~ 1.5 l) were collected using sterile plastic bottles. In the Metropolitan zone, samples were obtained from the midpoints of bridges to ensure representative sampling. Water filtration was performed on-site using 0.2 µm Sterivex filters and a portable Masterflex pump to preserve microbial integrity for downstream analysis.

DNA Extraction and Sequencing

Environmental DNA was isolated from the Sterivex filters using the NucleoMag® DNA/RNA Water Kit (Macherey–Nagel), following the manufacturer's instructions. DNA quality and yield were quantified using a Nanodrop spectrophotometer (Thermo Fisher Scientific, USA). Metagenomic DNA samples were diluted to a concentration of 5 ng/µl for sequencing. The multiplex 16S Barcoding Kit 1–24 SQK-16S024 (Oxford Nanopore, USA) was used following the manufacturer protocol for Flongle flow cells but adding double the suggested input DNA. This protocol involves the amplification of the full-length 16S rRNA gene with barcoded primers. PCR-amplified libraries for up to 24 samples were bead-purified and pooled in approximately equimolar amounts, followed by the addition of the rapid adaptor (RAP). Four sequencing pools were prepared and sequenced for 8 h on different Flongle flow cells using a MinION Mk1 C. Reads were basecalled using Guppy with its high accuracy model (HAC) and only those with an average quality score (Q) greater than 9 were retained for the analysis.

Bioinformatics and Data Analyses

Sequencing reads were demultiplexed and classified to a lowest common ancestor (LCA) using the FASTQ16S workflow incorporated in the EPI2ME Desktop Agent software. For the classification, reads were queried against the 16S + 18S rRNA database from NCBI (National Center for Biotechnology Information) with a minimum target coverage of 30%, minimum identity of 70%, and a maximum expectation



Fig. 1 Location of sampling points along the Rímac River. Upper Rimac: 1. Chicla, 2. San Mateo, 3. Tamboraque, 4. Hidroeléctrica, 5. Chacahuaro, 6. Matucana, 7. Surco. Middle Rimac: 8. Santa Eulalia,

9. Chaclacayo, 10. Huachipa. Lower Rimac (Metropolitan area): 11. Libertadores, 12. Nuevo, 13. Universitaria, 14. Faucett, 15. Gambetta

value (E) of 0.01. Alpha diversity indices (Chao1, Shannon, and Simpson) were estimated from the rarefied reads obtained using the Phyloseq package [15], and clustering dendrograms were generated. To test if alpha diversity indices differ significantly, we checked normality with the Shapiro–Wilk in the agricolae package [16]. For normal indices, we used Tukey’s post hoc tests; for non-normal indices, we used a Kruskal test. To estimate beta diversity, the OTU matrix was normalized using the variance stabilizing transformation (VST) method available in the DESeq2 package [17]. Subsequently, a multidimensional scaling (MDS) method utilizing a Euclidean similarity matrix was employed to compare beta diversity. Due to the observed differences in beta dispersion between groups, an Analysis of Similarity (ANOSIM) was employed, as it does not assume equal variance among groups [18]. The ampvis2 package [19] was employed to search for the most predominant ASV, generating heat maps of occurrences, and stack plots to view abundances. Differential abundance analysis was conducted utilizing Analysis of Compositions of Microbiomes with Bias Correction 2 (ANCOM-BC2) [20], focusing on

absolute abundance at the genus level across Metropolitan and Regional areas. p -values were adjusted using the Holm–Bonferroni method, and sensitivity analyses were incorporated to evaluate taxon significance.

Results

A total of 415,970 reads were sequenced across all samples, with an average of 7300 ± 4300 reads per sample. The read lengths averaged the expected full-length 16S rRNA gene length (~ 1.5 kb), and both Q scores and DNA throughput remained consistent and within the expected throughout the sequencing runs. Taxonomic analysis revealed that the most predominant class and phylum were Betaproteobacteria and Pseudomonadota, respectively. Additionally, 162 orders were identified, with Burkholderiales being the most abundant. At a finer taxonomic resolution, 3829 operational taxonomic units (OTUs) were detected.

Figure 2 illustrates the relative abundance of bacterial classes across each sampling point. Betaproteobacteria was

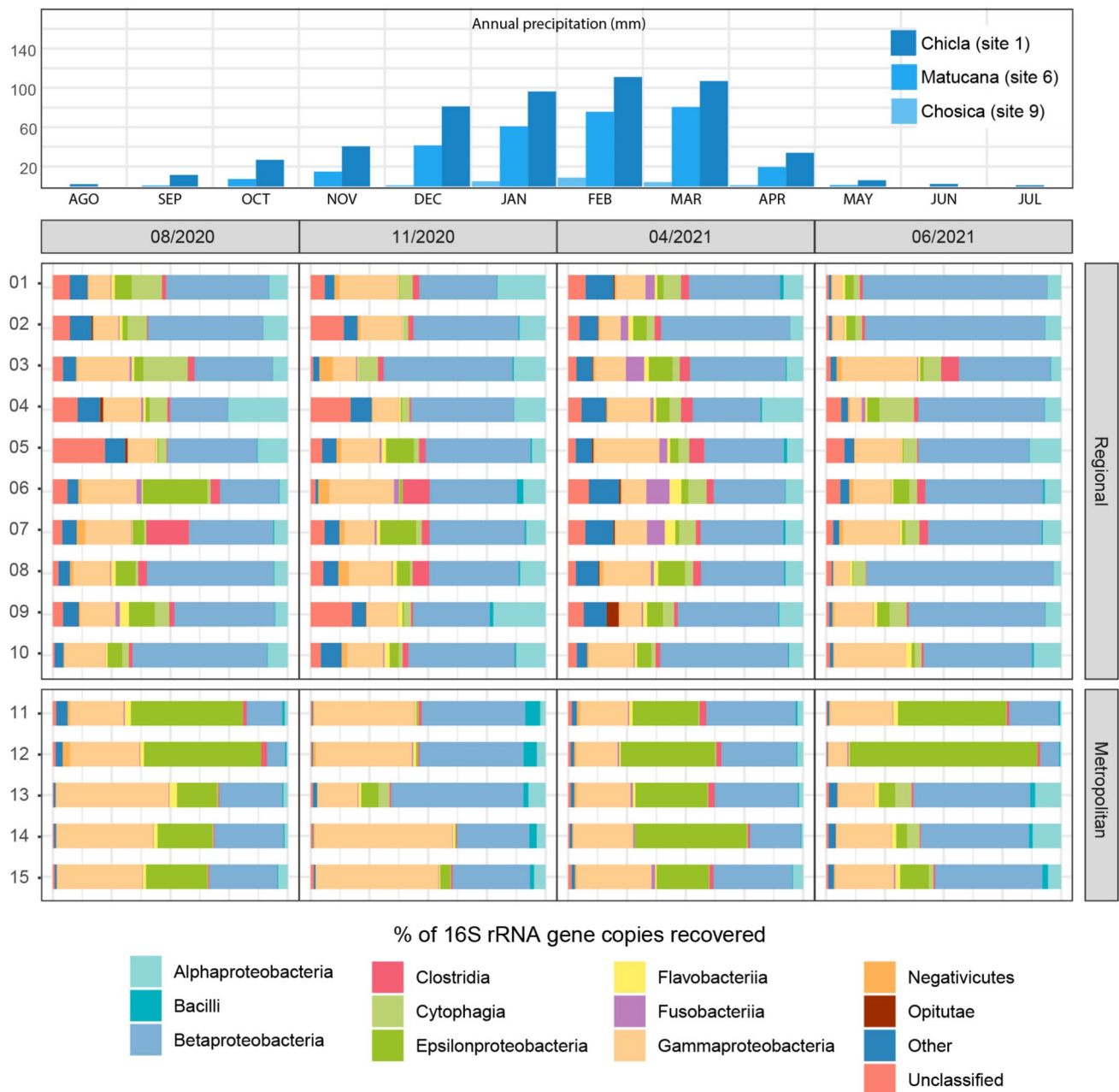


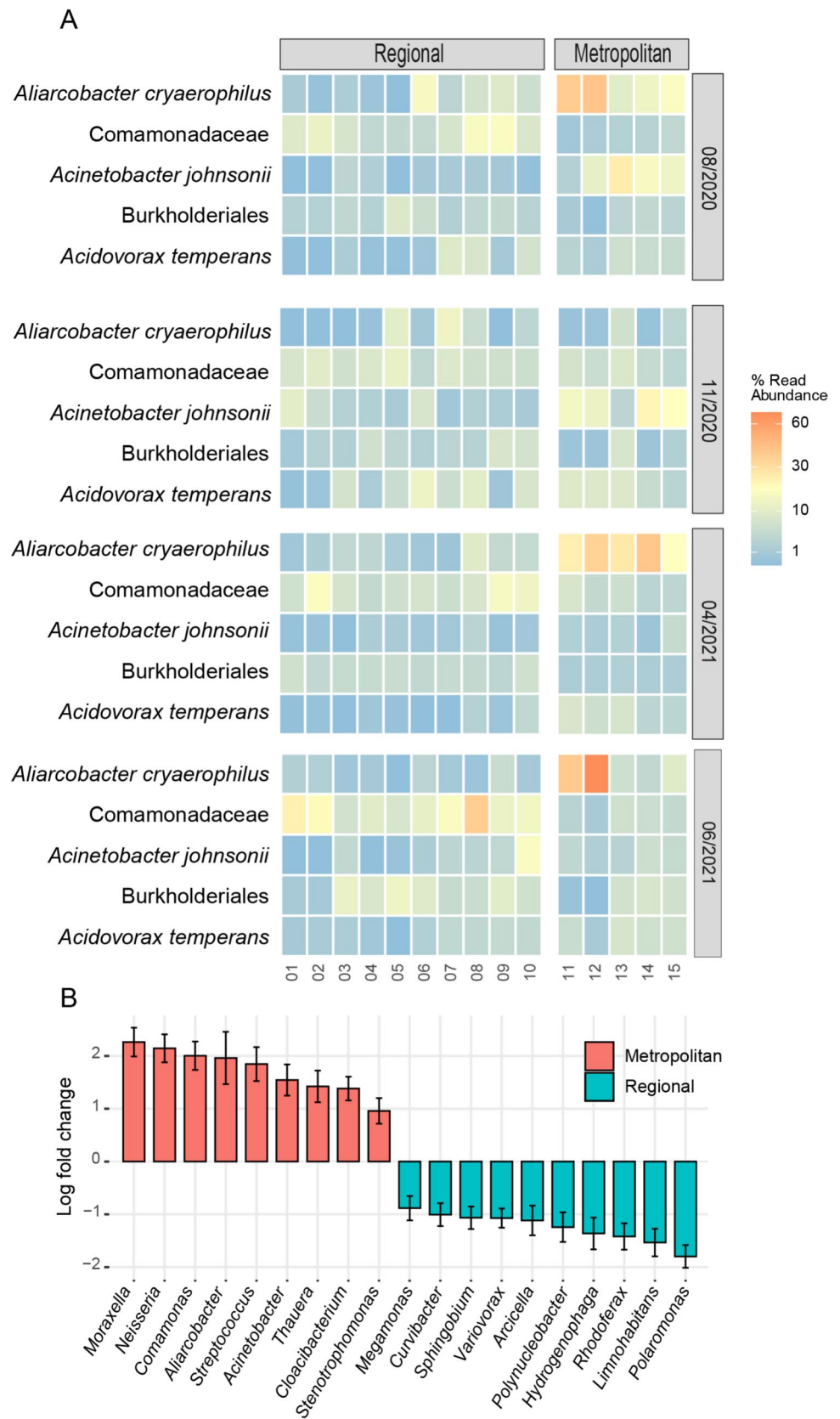
Fig. 2 Stacked bar diagram of the frequencies of the most common bacterial classes in the Rímac River, along its course. The sites are numbered from 1 to 15 according to Fig. 1. The histogram above

shows the annual precipitation (1981–2010) in the upper Rímac (Chicla, site 1, and Matucana, site 6) and the middle Rímac (Chosica, site 9). The lower part of the Rímac is a desert with scarce precipitation

the predominant class in most samples; however, shifts in microbial composition were observed in the Metropolitan zone, where Gammaproteobacteria and Epsilonproteobacteria were more prevalent. Figure 3 presents a heatmap of genus-level abundances, revealing that Comamonadaceae were the most abundant taxa in the upper and middle Rímac River throughout the year, while *Aliarcobacter cryaerophilus* and *Acinetobacter johnsonii* dominated the Metropolitan area.

Hierarchical clustering analysis of microbial communities by sampling event is depicted in Fig. 4. Two distinct clusters were identified, with one primarily composed of samples from the Metropolitan area, indicating high similarity in microbial composition within this region. Similarly, multi-dimensional scaling (MDS) analysis (Fig. 4) demonstrated a clear separation between Metropolitan samples and those from the rest of the basin (Regional), both when analyzed by season and as a whole. ANOSIM indicated a significant

Fig. 3 **A** Heatmap of the abundance of the ten most frequent bacterial taxa in the Rímac River. In the x-axis, the sites are numbered from 1 to 15 according to Fig. 1. **B** Significantly different abundant bacterial genus taxa according to ANCOM-BC2 ($p < 0.05$). Bars indicate log₂-fold differences in abundance between Regional (negative) and Metropolitan (positive) areas



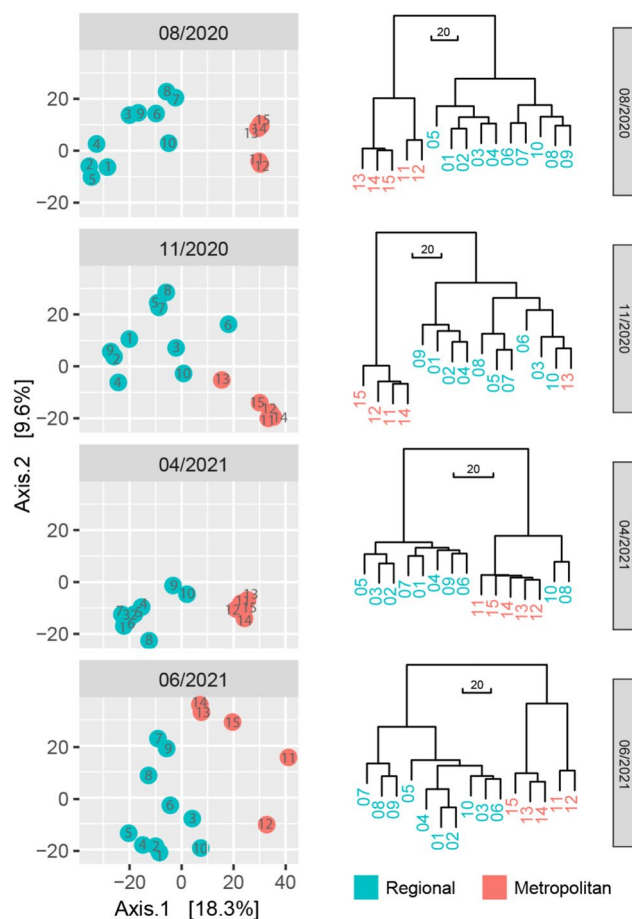


Fig. 4 Multidimensional scaling in all samples with respect to regions and time of year. Dendrograms show the clustering of microbial composition in the 4 samplings carried out along the Rímac River. The sites are numbered from 1 to 15 according to Fig. 1

separation between the Regional and Metropolitan areas ($R = 0.48$, $p = 0.001$). Additionally, ANOSIM showed no difference between the upper basin (points 1–7) and the middle basin (points 8–10) ($p < 0.21$) but did indicate differences between the lower part of the basin (Metropolitan area, points 11–15) and the rest of the basin ($p < 0.001$). Conversely, there was no significant difference between the dry and rainy seasons ($R = 0.05$, $p = 0.025$). When considering only samples from the Regional part by month of collection, all comparisons were significant ($p < 0.05$). On the other hand, samples from the Metropolitan part do not differ statistically from each other ($p > 0.01$).

Alpha diversity indices, as shown in Fig. 5, revealed notable spatial and temporal variations. The indices indicated significant differences ($p < 0.05$) during the first three sampling events, with lower richness, diversity, and evenness in the Metropolitan region compared to the Regional area. This trend was reversed in the fourth sampling, suggesting dynamic shifts in microbial community structure over time,

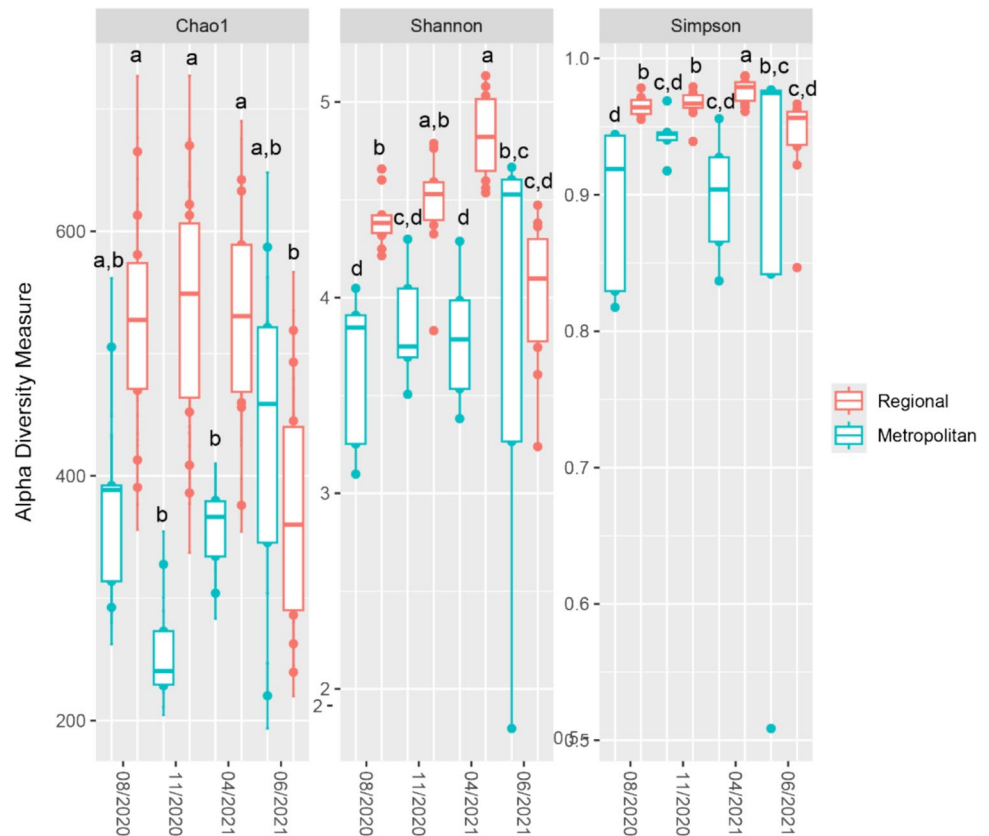
although this difference was not statistically significant. These findings underscore the influence of anthropogenic activities on microbial diversity in the Metropolitan zone and highlight the importance of temporal considerations in ecological assessments.

Discussion

In contemporary environmental microbiology, it is imperative not only to identify bacterial communities within aquatic ecosystems but also to elucidate the spatial and temporal dynamics governing their composition [21]. This study, encompassing a year-long sampling effort, provides the most comprehensive characterization to date of the bacterial community diversity along the Rímac River, utilizing a 16S rRNA gene metabarcoding approach. We systematically compared microbial assemblages across seasonal variations (dry and rainy seasons) and spatial gradients (Regional vs. Metropolitan zones). Sampling locations included those previously examined by Romero et al. [5], supplemented by two additional sites in the mid-basin region.

Both hierarchical clustering and multidimensional scaling (MDS) analyses revealed distinct subdivisions within the microbial communities of the Rímac River (Fig. 4). Hierarchical clustering grouped Metropolitan samples into a single cluster, indicative of compositional homogeneity, whereas Regional samples formed separate clusters, reflecting greater variability (Fig. 4). Similarly, MDS and ANOSIM analyses demonstrated a significant separation between Metropolitan and Regional samples across all sampling events (Fig. 4). Notably, Regional samples exhibited higher dispersion and diversity indices (Fig. 5), while Metropolitan samples were significantly more uniform and less diverse, with the exception of June 2021. The dispersion observed in Regional samples may be attributable to altitudinal gradients (0–4500 m above sea level), as evidenced by MDS. Specifically, upper Rímac sites (localities 1–5) clustered more closely together than mid-Rímac sites (localities 6–9), particularly during the dry season (Fig. 4), although no statistical difference in bacterial communities between the upper and the middle basin was observed. Additionally, no significant differences were found according to seasonal variations between rainy and dry periods. However, when samples from the Metropolitan area are excluded from the analysis, significant monthly differences in the Regional area are detected. It could be observed in the samples collected in April 2021 within the Regional zone that reduced dispersion and limited overlap with those from November 2020 likely reflect the influence of peak rainfall occurring in February and March. Our results indicate that Lima exerts a significant impact over the microbial communities in the Rímac River, overshadowing other factors, such as seasonal variations. Metropolitan

Fig. 5 Alpha, Chao1 (richness), Shannon (diversity), and Simpson (evenness) diversity indices of the four samples campaigns. Letters above boxplots represent significant differences ($p < 0.05$) between average values of each sub-basin



areas significantly alter river systems, leading to notable changes in microbial communities. This phenomenon has been observed in various countries, including Brazil [6] and Nepal [4]. This difference has been linked to bacterial genera that are associated with human pathogens originating from fecal matter [4–7, 22].

Betaproteobacteria, a class within the phylum Pseudomonadota, emerged as the most abundant taxon across most samples. This group [21] is commonly found in aquatic environments, including rivers [23, 24] and lakes [25, 26]. Interestingly, Romero et al. [5] reported Bacteroidia, particularly *Flavobacterium*, as the dominant class, whereas we observed *Flavobacteriia* at lower abundances. This discrepancy likely stems from the use of different sequencing platforms, as biases inherent to each technology have been reported [27, 28]. Despite these variations, both studies consistently identified distinct microbial patterns between Regional and Metropolitan zones.

The shift in bacterial composition within the Metropolitan zone is characterized by increased abundances of Gammaproteobacteria and Epsilonproteobacteria (Fig. 2). Nine genera, including *Aliarcobacter* and *Acinetobacter*, were found to be differentially abundant in the Metropolitan area (Fig. 3B). Most of these genera have previously been identified in rivers and are often linked to water pollution and sewage impact [29–33], supporting the evidence of human

impact in the lower part of the Rímac River. Notably, *Aliarcobacter cryaerophilus* (Epsilonproteobacteria), a genus indicative of fecal contamination and a potential public health risk [6], was the most prevalent species in three of the four sampling campaigns (Fig. 3A). This finding aligns with Romero et al. [5], who identified *Arcobacter cryaerophilus* (synonymous with *Aliarcobacter cryaerophilus*) as the dominant species. Classified as a serious hazard by the International Commission on Microbiological Specifications for Food (ICMSF, 2002), *A. cryaerophilus* is associated with intestinal and extraintestinal infections, including bacteremia, peritonitis, and severe diarrhea [34]. Its resilience in water and wastewater systems [35], coupled with antibiotic resistance and potential for horizontal gene transfer [36], underscores its public health significance. Despite its prevalence in the Rímac River, reports of *Aliarcobacter*-related infections in Peru remain limited, with only one study documenting its presence in children with diarrhea [37]. The second most abundant species in the Metropolitan zone was *Acinetobacter johnsonii* (Gammaproteobacteria), except in November 2020, when it was the most dominant. *Acinetobacter* spp. are ubiquitous in natural environments, including soil, water, and sewage [38], with *A. johnsonii* frequently isolated from human skin and feces [38].

This study highlights the pronounced and consistent differences in microbial communities between the

Metropolitan and Regional zones, driven by reduced diversity and the prevalence of pathogenic bacteria (i.e., *Aliarcobacter cryaerophilus*) in the Metropolitan area. These findings underscore the profound anthropogenic impact of Lima on the Rímac River. Municipal wastewater is the main source of pollution, accounting for 62% of the total [39]. This includes 450 drainage tubes that are discharged directly into the river without any treatment [39]. Most of them are in Lurigancho (Chosica), near point 11 and 12, the point with more presence of *Aliarcobacter cryaerophilus*. Our findings underscore the necessity of establishing a national plan to enhance the water quality of the Rimac River, which includes constructing new wastewater treatment plants (WWTP) and augmenting the capacity of existing facilities. Also, a robust monitoring program is essential, incorporating expanded sampling efforts, particularly during the rainy season, which was herein under-represented due to COVID-19-related restrictions. Such initiatives are critical for informing conservation strategies and mitigating public health risks associated with microbial contamination in this vital water resource.

Author Contribution All authors contributed to the study conception and design. TS, GO, and RLT contributed to material preparation and data collection. TS, RLT, and JLR were responsible for analysis. The first draft of the manuscript was written by TS. JLR and GO contributed to the revision and editing of the manuscript. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data Availability Sequence data that support the findings of this study have been deposited in NCBI BioProject database (PRJNA1223955).

Declarations

Competing interests The authors declare no competing interests.

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